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Review

A review on polysaccharide biosynthesis in *Cordyceps militaris*

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Abstract

Cordyceps militaris (*C. militaris*) is an edible parasitic fungus with medicinal properties. Its bioactive polysaccharides are structurally diverse and exhibit various metabolic and biological activities, including antitumor, hypoglycemic, antioxidant, hypolipidemic, anti-inflammatory, immunostimulatory, and anti-atherosclerotic effects. These properties make *C. militaris*-derived polysaccharides a promising candidate for future development. Recent advancements in microbial fermentation technology have enabled successful laboratory cultivation and extraction of these polysaccharides. These polysaccharides are structurally diverse and exhibit various biological activities, such as immunostimulatory, antioxidant, antitumor, hypolipidemic, and anti-atherosclerotic effects. This review aims to summarize the structure and production mechanisms of polysaccharides from *C. militaris*, covering extraction methods, key genes and pathways involved in biosynthesis, and fermentation factors that influence yield and activity. Furthermore, the future potential and challenges of utilizing polysaccharides in the development of health foods and pharmaceuticals are addressed. This review serves as a valuable reference in the fields of food and medicine, and provides a theoretical foundation for the study of polysaccharides.

Introduction

Cordyceps militaris (*C. militaris*), an entomopathogenic fungus belonging to the genus *Cordyceps*, has gained significant attention because of its chemical composition and pharmacological properties that are similar to those of *Cordyceps sinensis* [1,2]. The bioactive components of *C. militaris* include cordycepin acid, mannitol, alkaloids, cordycepin, vitamins B1 and B2, and polysaccharides [3]. Polysaccharides in *C. militaris*, constituting 3–8 % of their total weight, are the primary active compounds and demonstrate blood sugar-lowering, antioxidant, immune-boosting, antitumor, antibacterial, and anti-inflammatory effects [4]. There is a consistently high market demand for *C. militaris*. However, the stringent environmental requirements of *C. militaris* for its natural growth have contributed to an annual decline in its population. Wild-type *C. militaris* parasitizes insect pupae, including lepidopterans [5]. *C. militaris* hyphae grow and die normally within the pupae, and then emerge from the head, chest, or abdomen of the pupae [5]. Wild-type *C. militaris* displays an orange-yellow coloration (Fig. 1a). *C. militaris* has been successfully cultivated in the laboratory using potato as a solid culture medium, resulting in a white and fluffy appearance (Fig. 1b) [6]. In recent years, liquid fermentation media (Fig. 1c) have attracted increasing attention owing to their short cultivation time and consistent yield [7]. Researchers have isolated and cultivated multiple strains of *C. militaris* that produce active substances with pharmacological effects similar to those found in natural fungi [8]. Therefore, optimization of the artificial cultivation of *C. militaris* using fermentation technology is of utmost importance.

Selenium-enriched polysaccharides (ISPS) can be produced by cultivating *C. militaris* in selenium-rich fermentation medium [9]. This method improves *C. militaris* productivity and boosts polysaccharide production without affecting its activity [10]. The production of polysaccharides by *C. militaris* in a fermentation medium is influenced by various factors, such as carbon and nitrogen sources, C/N ratio, inorganic salts, trace elements, cultivation temperature, rotational speed, oxygen content, light conditions, pH, and cultivation time [11]. Despite efforts to optimize these factors, the current production of polysaccharides from *C. militaris* is inadequate to meet the market demand. The manipulation of specific genes and enzymes involved in polysaccharide biosynthesis in *C. militaris* has been shown to significantly enhance polysaccharide production [12]. Therefore, comprehensive analysis of polysaccharide biosynthetic pathways holds promise for increasing their production.

Recent studies have extensively focused on the biosynthesis of fungal polysaccharides, which involves various processes, including the synthesis of nucleotide sugar precursors, initial synthesis reactions, and subsequent processes such as extension, flipping, folding, and polymerization of oligosaccharide repeat units [13,14]. The entire genome of *C. militaris* 01 has been successfully sequenced [15], and gene annotations and essential enzymes for polysaccharide synthesis have been consistently identified. Key enzymes in the biosynthesis of nucleotide sugar precursors are glucokinase (Gk), UDP-glucose pyrophosphorylase (Ugp), phosphoglucose mutase (Pgm), and UDP-glucose 6-dehydrogenase (Ugdh) [16]. Therefore, enhancing polysaccharide production in *C. militaris* through gene editing is a prominent research area.

C. militaris-derived polysaccharides have gained significant attention for their potential health benefits and are currently the focus of quality control and development initiatives. Most reviews have primarily focused on the isolation, extraction, factors affecting activity, and biological functions of these polysaccharides. Miao et al. investigated the structure, characteristics, and signaling pathways regulated by *C. militaris* polysaccharides [17]. Zhang presented an overview of the advancements in extraction, separation, structure, and biological activity of these polysaccharides [18]. Wu et al. discussed the recent progress in the chemical characteristics as well as biosynthetic gene clusters and pathways of various bioactive components of *C. militaris*, such as cordycepin, D-mannitol, N6-(2-hydroxyethyl)-adenosine (HEA), and polysaccharides [19]. However, studies on the biosynthetic pathways of *C. militaris* polysaccharides are limited. Yang et al. conducted a comprehensive review of the biosynthetic pathway and polysaccharide production in all *Cordyceps* genera [20]. However, the influence of fermentation factors on the extraction, structure, and yield of *C. militaris* polysaccharides is not yet completely understood. Furthermore, the relationship between key genes involved in polysaccharide synthesis and yield in *C. militaris* remains unclear. The purpose of this review is to provide a comprehensive overview of the extraction methods, structural characteristics, synthesis pathways, and key genes associated with polysaccharides in *C. militaris*. This review also summarizes the fermentation factors of *C. militaris* and provides a theoretical basis for future research on polysaccharides. Relevant literature from 2016 to the present was obtained from the US National Library of Medicine (PubMed) using the keywords “*Cordyceps militaris* polysaccharide,” “structure,” “extraction,” “activity,” “polysaccharide biosynthesis,” and “fermentation.”

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Extraction of polysaccharides from *C. militaris*

Although *C. militaris* has an extensive natural growth range, its low growth density can adversely affect the ecological environment owing to excessive excavation [21]. Therefore, researchers are currently cultivating *C. militaris* in laboratory settings using liquid or solid-state fermentation. Liquid fermentation involves growing the fungus in a liquid medium, such as a broth or nutrient solution, under controlled temperature, pH, and oxygen supply. In contrast, solid-state fermentation

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Chemical structures of *C. militaris* polysaccharides

Various polysaccharides have been extracted from *C. militaris*. The characteristics and biological activities of these polysaccharides are affected by their molecular weight and distribution. The value and quality of polysaccharides depend on their molecular weight, size, and distribution. Various techniques, including gel filtration, viscosity measurement, osmotic pressure, sedimentation, methylation analysis, high-performance gel permeation chromatography (HPGPC), and gel chromatography ...

Analysis of biosynthetic pathways and key genes of polysaccharides

The function of polysaccharides is influenced by their monosaccharide composition, glycosidic bonds, and degree of polymerization [69]. *C. militaris* polysaccharides consist primarily of glucans linked by glycosidic bonds, including (1 → 3)- and (1 → 6)- β or (1 → 3)-, (1 → 4)-, and (1 → 6)- α -glucans [70,71]. The biosynthesis of β -glucan involves the synthesis of the precursor UDP-glucose. It also involves polymerizing, extending, and branching β -glucan chains. This process relies on the ...

Factors affecting the yield and activity of *C. militaris* polysaccharides

The bioactivities of polysaccharides in *C. militaris* have been reviewed by Yang et al. [18], and the fermentation conditions of *C. militaris*-derived polysaccharides have been reported by Miao et al. [17]. Here, we have summarized in detail the factors that affect the biological activity and yield of *C. militaris* polysaccharides. Several factors limit the biological activity and yield of *C. militaris* polysaccharides. Local environmental factors, such as soil, water, weather, and other natural ...

Conclusions and future directions

Various bioactive compounds, including cordycepin, adenosine, ergosterol, myriocin, and polysaccharides, have been isolated and characterized from *C. militaris*. *C. militaris* polysaccharides have various functions, such as reducing blood sugar, possessing anticancer properties, preventing arteriosclerosis, enhancing the immune system, and acting as antioxidants. Therefore, research on polysaccharides has gained momentum in recent years. Most reviews have focused on the synthesis of ...

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CRediT authorship contribution statement

Na Wu: Writing – original draft, Funding acquisition, Conceptualization. **Xiaodong Ge:** Writing – original draft, Funding acquisition, Conceptualization. **Xuemei Yin:** Writing – original draft, Funding acquisition, Conceptualization. **Lei Yang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Ligen Chen:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Rong Shao:** Writing – review & editing, Supervision, Funding acquisition, ...

Declaration of competing interest

The authors confirm that we have no conflicts of interest with respect to the work described in this manuscript. ...

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Not applicable. ...

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References (124)

G.-H. Sung *et al.*

[Phylogenetic classification of *Cordyceps* and the clavicipitaceous fungi](#)

Studies in Mycology (2007)

F.R. Smiderle *et al.*

[Isolation and chemical characterization of a glucogalactomannan of the medicinal mushroom *Cordyceps militaris*](#)

Carbohydrate Polymers (2013)

W. Wang *et al.*

[Manipulation of ginsenoside heterogeneity in cell cultures of *Panax notoginseng* by addition of jasmonates](#)

Journal of Bioscience and Bioengineering (2002)

V.Y. Kryukov *et al.*

[Temperature adaptations of *Cordyceps militaris*, impact of host thermal biology and immunity on mycosis development](#)

Fungal Ecology (2018)

J. Zhang *et al.*

Advance in *Cordyceps militaris* (Linn) Link polysaccharides: isolation, structure, and bioactivities: A review

International Journal of Biological Macromolecules (2019)

X. Chen *et al.*

Structural analysis and antioxidant activities of polysaccharides from cultured *Cordyceps militaris*

International Journal of Biological Macromolecules (2013)

J.S. Lee *et al.*

Mechanism of macrophage activation induced by polysaccharide from *Cordyceps militaris* culture broth

Carbohydrate Polymers (2015)

Z.-B. Wang *et al.*

Effect of extraction media on preliminary characterizations and antioxidant activities of *Phellinus linteus* polysaccharides

Carbohydrate Polymers (2014)

C. Wang *et al.*

Optimization for the extraction of polysaccharides from *Gentiana scabra* Bunge and their antioxidant in vitro and anti-tumor activity in vivo

Journal of the Taiwan Institute of Chemical Engineers (2014)

Z.-y. Zhu

Dong F, Liu X, Lv Q, YingYang, Liu F, Chen L, Wang T, Wang Z, Zhang Y: Effects of extraction methods on the yield, chemical structure and anti-tumor activity of polysaccharides from *Cordyceps gunnii* mycelia

Carbohydrate Polymers (2016)



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Cited by (27)

Refined regulation of polysaccharide biosynthesis in edible and medicinal fungi: From pathways to production

2025, Carbohydrate Polymers

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Structural characterization and hypoglycemic activity of *Cordyceps sinensis* polysaccharide (CSP-3)

2024, Food Bioscience

Citation Excerpt :

...Li et al. demonstrated that $\rightarrow 4$)- α -D-Glcp (1 $\rightarrow$ -linked dextran is specific to *Cordyceps* and can serve as a structural marker (Li, But, et al., 2021). However, this does not represent a universal structure paradigm for *Cordyceps* (Miao et al., 2022; Wu et al., 2024). The structural diversity of heteropolysaccharides contributes to their wide array of functionalities and significantly influences their biological activity....

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Citation Excerpt :

...Consequently, the production of carotenoids may involve numerous inducible elements, which could boost the development of DNA elements in synthetic biology. The biosynthetic pathway for CPs comprises several steps, although the complete pathway has not been fully elucidated (Wu et al., 2024). The pathway begins with the synthesis of nucleotide-activated sugars, which serve as substrates for the initial synthesis of CPs....

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...Different glycosyltransferases (GTs) sequentially transfer monosaccharides from sugar nucleotide donors to lipid carriers, assembling them into repeat units. These repeat units are then aggregated and exported (Mohammadi et al., 2011; Wu et al., 2024). Phosphoglucomutase (Pgm), UDPG-pyrophosphorylase (Ugp), phosphoglucomutase (Pgi), UDPG-dehydrogenase (Upgh), and glucokinase (Gk) are some of the enzymes that are crucial for making the building blocks that are needed for *C. militaris* polysaccharide synthesis....

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